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Technologies for reducing event notifications in telematics systems

Abstract

Technologies for reducing event notifications in telematics systems includes a telematics server, a telematics device located in a vehicle, and a notification device. The telematics server receives telematics data from the telematics device and analyzes the telematics data based on an event model to determine whether an unexpected event has occurred. The event model includes a plurality of rules that define expected events related to the vehicle based on historical telematics data. The server generates a notification in response to a determination that the unexpected event has occurred and transmits the notification to the notification device, which may generate a notification to an end user. The event model may be generated and updated by the telematics server using an unsupervised and/or supervised machine learning algorithm based on the telematics data.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6167448	12/1999	Hemphill et al.	N/A	N/A
9331926	12/2015	Oh et al.	N/A	N/A
9977579	12/2017	Yang et al.	N/A	N/A
10328204	12/2018	Davis et al.	N/A	N/A
10347373	12/2018	Brill et al.	N/A	N/A
10394546	12/2018	Hu et al.	N/A	N/A
10710456	12/2019	Mueller et al.	N/A	N/A
11295140	12/2021	Wang et al.	N/A	N/A
2003/0065788	12/2002	Salomaki	N/A	N/A
2004/0098491	12/2003	Costa-Requena et al.	N/A	N/A
2005/0135240	12/2004	Ozugur	N/A	N/A
2005/0154698	12/2004	Ikezawa et al.	N/A	N/A
2006/0149816	12/2005	Cadiz et al.	N/A	N/A
2006/0240803	12/2005	Valeriano et al.	N/A	N/A
2006/0286993	12/2005	Zie et al.	N/A	N/A
2018/0124213	12/2017	Wood	N/A	H04L 47/50
2020/0363914	12/2019	Dascola et al.	N/A	N/A
2023/0260046	12/2022	Estes	705/4	G06Q 10/06315

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20060070419	12/2005	KR	N/A
2007069992	12/2006	WO	N/A

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to telematics and associated telematics devices and, more particularly, to telematics systems configured for providing notifications to end users.

BACKGROUND

(2) Telematics involves the integrated use of telecommunications and informatics, particularly for

application in vehicles. Telematics functionality may include, but is not limited to, recording vehicle information, emergency warning systems, navigation functionality, safety warnings, and automated driving assistance. Vehicle telematics devices may process and transmit telematics data and other messages generated by a vehicle and connected devices. For example, the telematics data may be produced by sensors, controllers, and/or other peripheral devices located on the vehicle and/or within the telematics device itself.

(3) In some telematics systems, the vehicle telematics devices may transmit the locally collected telematics data to a telematics server. In typical telematics systems, especially those related to fleet operations, Transportation & Logistics (T&L), and Connected Vehicles, the telematics server may continually or periodically transmit notifications to an end user of the system, such as a fleet manager, a driver of the associated vehicle, or a customer of the service (e.g., a parent of a monitored child) based on the received telematics data to keep the end user informed and the telematics device up-to-date. However, such reoccurring notifications can consume a large amount of bandwidth and become excessive and unmanageable for the end user.

SUMMARY

(4) According to an aspect of the present disclosure, a telematics server for providing notifications to a telematics device of a vehicle may include an event model, a device interface, and a notification manager. The event model may include a plurality of rules that each define an expected event based on historical telematics data of the vehicle. The device interface may be configured to receive telematics data from the telematics device, and the telematics data may define one or more characteristics of the vehicle. The notification manager may be configured to analyze the telematics data using the event model to determine whether an unexpected event has occurred and, in response to a determination that the unexpected event has occurred, generate a notification. The device interface may be further configured to transmit the notification to a notification device.

(5) In some embodiments, the notification manager may be configured to update the event model based on the telematics data received from the telematics device or based on feedback received from the notification device using a machine learning engine. Additionally, in some embodiments, the notification may instruct the notification device to generate an alert. Additionally or alternatively, the notification may be configured to provide information related to the unexpected event to a user of the notification device. In some embodiments, the telematics server may further include a model manager including a machine learning engine configured to generate the event model based on the historical telematics data using unsupervised learning.

(6) Additionally, in some embodiments, to analyze the telematics data may include to analyze the telematics data based on a temporal-based rule of the event model. In such embodiments, the temporal-based rule may define an expected temporal event related to the vehicle based on the historical telematics data. Additionally, in some embodiments, to analyze the telematics data may include to analyze the telematics data based on a location-based rule of the event model. In such embodiments, the location-based rule may define an expected location of the vehicle based on historical telematics data. Additionally, in some embodiments, to analyze the telematics data may include to analyze the telematics data based on a historical-based rule of the event model. In such embodiments, the historical based rule may define an expected activity related to the vehicle based on the historical telematics data. Additionally, in some embodiments, to analyze the telematics data may include to analyze the telematics data based on a sequential-based rule of the event model. In such embodiments, the sequential-based rule may define an expected event related to the vehicle given an occurrence of a prior event related to the vehicle. Additionally, in some embodiments, to analyze the telematics data may include to analyze the telematics data based on a relational-based rule of the event model. In such embodiments, the relational-based rule may define an expected characteristic of the vehicle given a state of another characteristic of the vehicle.

(7) According to another aspect of the present disclosure, a method for providing notifications to a telematics device of a vehicle may include receiving, by a telematics server, telematics data from

the telematics device; analyzing, by the telematics server, the telematics data using an event model managed by the telematics server to determine whether an unexpected event has occurred; generating, by the telematics server and in response to a determination that the unexpected event has occurred, a notification; and transmitting, by the telematics server, the notification to a notification device. The telematics data may define one or more characteristics of the vehicle, and the event model may include a plurality of rules that each define an expected event based on historical telematics data.

(8) In such embodiments, the method may further include updating, by the telematics server, the event model based on the telematics data received from the telematics device or based on feedback received from the notification device using a machine learning engine. Additionally, in some embodiments, the notification may instruct the notification device to generate an alert or is configured to provide information related to the unexpected event to a user of the notification device.

(9) Additionally, in some embodiments, analyzing the telematics data may include analyzing the telematics data based on a temporal-based rule of the event model. In such embodiments, the temporal-based rule may define an expected temporal event related to the vehicle based on the historical telematics data. Additionally, in some embodiments, analyzing the telematics data may include analyzing the telematics data based on a location-based rule of the event model. In such embodiments, the location-based rule may define an expected location of the vehicle based on historical telematics data. Additionally, in some embodiments, analyzing the telematics data may include analyzing the telematics data based on a historical-based rule of the event model. In such embodiments, the historical based rule may define an expected activity related to the vehicle based on the historical telematics data. Additionally, in some embodiments, analyzing the telematics data may include analyzing the telematics data based on a sequential-based rule of the event model. In such embodiments, the sequential-based rule may define an expected event related to the vehicle given an occurrence of a prior event related to the vehicle. Additionally, in some embodiments, analyzing the telematics data may include analyzing the telematics data based on a relational-based rule of the event model. In such embodiments, the relational-based rule may define an expected characteristic of the vehicle given a state of another characteristic of the vehicle.

(10) According to yet another aspect of the present disclosure, one or more non-transitory, computer-readable storage media comprising a plurality of instructions that, in response to execution, cause a telematics server to receive telematics data from the telematics device, analyze the telematics data using an event model managed by the telematics server to determine whether an unexpected event has occurred, generate, in response to a determination that the unexpected event has occurred, a notification, and transmit the notification to a notification device. In some embodiments, the telematics data defines one or more characteristics of the vehicle, and the event model includes a plurality of rules that each define an expected event based on historical telematics data. Additionally in some embodiments, the plurality of instructions, in response to execution, may further cause the telematics server to update the event model based on the telematics data received from the telematics device or based on feedback received from the notification device using a machine learning engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The concepts described herein are illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. Where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.

- (2) FIG. 1 is a simplified block diagram of at least one embodiment of a telematics system for reducing event notifications to a telematics device;
- (3) FIG. 2 is a simplified block diagram of various environments that may be established by the telematics system of FIG. 1;
- (4) FIG. 3 is a simplified flow diagram of at least one embodiment of a method for managing telematics data that may be executed by a vehicle telematics device of the telematics system of FIGS. 1 and 2;
- (5) FIG. 4 is a simplified flow diagram of at least one embodiment of a method for constructing an event model for analyzing telematics data that may be executed by a telematics server of the telematics system of FIGS. 1 and 2;
- (6) FIG. 5 is a simplified flow diagram of at least one embodiment of a method for providing notifications to a notification device that may be executed by the telematics server of the telematics system of FIGS. 1 and 2;
- (7) FIG. 6 is a simplified flow diagram of at least one embodiment of a method for managing notifications received from the telematics server that may be executed by the notification device of the telematics system of FIGS. 1 and 2; and
- (8) FIG. 7 is a simplified flow diagram of at least one additional embodiment of a method for managing telematics data that may be executed by the vehicle telematics device of the telematics system of FIGS. 1 and 2.

DETAILED DESCRIPTION OF THE DRAWINGS

- (9) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.
- (10) References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (B and C); or (A, B, and C).
- (11) The disclosed embodiments may be implemented, in some cases, in hardware, firmware, software, or any combination thereof. The disclosed embodiments may also be implemented as instructions carried by or stored on a transitory or non-transitory machine-readable (e.g., computer-readable) storage medium, which may be read and executed by one or more processors. A machine-readable storage medium may be embodied as any storage device, mechanism, or other physical structure for storing or transmitting information in a form readable by a machine (e.g., a volatile or non-volatile memory, a media disc, or other media device).
- (12) In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with

other features.

(13) Referring now to FIG. 1, an illustrative telematics system **100** for providing notifications to a telematics device includes a telematics server **102**, a telematics device **104** installed in a vehicle **106**, and a notification device **108**. Each of the telematics device **104** and the notification device **108** are in communication with the telematics server **102** over a network **110**. In use, as described in more detail below, the telematics server **102** receives telematics data from the telematics device **104** via the network **110**, analyzes the telematics data based on an event model (e.g., a machine learning model) to determine whether an unexpected event has occurred as indicated by the telematics data, and transmits a notification to the notification device **108** via the network **110** if the occurrence of an unexpected event has been detected. As such, the overall bandwidth of notifications between the telematics server **102** and the notification device **108** may be reduced by transmitting only notifications regarding unexpected events (e.g., the vehicle **106** is in an unexpected location, the vehicle **106** has failed to arrive at a location at an expected time, the vehicle **106** is traveling an unexpected rate of speed, etc.). In some embodiments as discussed in more detail below, the event model may be constructed, updated, and/or maintained using a machine learning algorithm based on received telematics data such that the expected behavior of the vehicle **106** can be more accurately determined over time.

(14) The telematics server **102** may be embodied as any type of computation or computer device capable of performing the functions described herein, including, without limitation, a server, a rack-mounted server, a blade server, a workstation, a network appliance, a web appliance, a desktop computer, a laptop computer, a tablet computer, a smartphone, a consumer electronic device, a distributed computing system, and/or a multiprocessor system. Additionally, in some embodiments, the telematics server **102** may be embodied as a “virtual server” formed from multiple computing devices distributed across the network **110** and operating in a public or private cloud. Accordingly, although the telematics server **102** is illustrated in FIG. 1 as embodied as a single computing device, it should be appreciated that the telematics server **102** may be embodied as multiple devices cooperating together to facilitate the functionality described below.

(15) As shown in FIG. 1, the illustrative telematics server **102** includes compute circuitry **120**, an I/O subsystem **126**, a data storage device **128**, and a communication subsystem **130**. Of course, the telematics server **102** may include other or additional components, such as those commonly found in a typical server or other computer (e.g., various input/output devices), in other embodiments. Additionally, in some embodiments, one or more of the illustrative components may be incorporated in, or otherwise form a portion of, another component.

(16) The compute circuitry **120** may be embodied as any type of device or collection of devices capable of performing various compute functions. In some embodiments, the compute circuitry **120** may be embodied as a single device such as an integrated circuit, an embedded system, a field-programmable-array (FPGA), a system-on-a-chip (SOC), or other integrated system or device. Additionally, in some embodiments, the compute circuitry **120** includes or is embodied as a processor **122** and memory **124**. The processor **122** may be embodied as any type of processor capable of performing the functions described herein. For example, the processor **122** may be embodied as a single or multi-core processor(s), digital signal processor, microcontroller, or other processor or processing/controlling circuit. Similarly, the memory **124** may be embodied as any type of volatile or non-volatile memory or data storage capable of performing the functions described herein. In operation, the memory **124** may store various data and software used during operation of the telematics server **102** such as operating systems, applications, programs, libraries, and drivers.

(17) The compute circuitry **120** is communicatively coupled to other components of the telematics server **102** via the I/O subsystem **126**, which may be embodied as circuitry and/or components to facilitate input/output operations with compute circuitry **120** (e.g., with the processor **122** and/or memory **124**) and other components of the telematics server **102**. For example, the I/O subsystem

126 may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations. In some embodiments, the I/O subsystem **126** may be incorporated, along with the processor **122**, the memory **124**, and other components of the telematics server **102**, into the compute circuitry **120**.

(18) The data storage device **128** may be embodied as any type of device or devices configured for short-term or long-term storage of data such as, for example, memory devices and circuits, memory cards, hard disk drives, solid-state drives, or other data storage devices. The communication subsystem **130** of the telematics server **102** may be embodied as any communication circuit, device, or collection thereof, capable of enabling communications between the telematics server **102**, the telematics device **104**, the notification device **108**, and/or other remote devices. The communication subsystem **130** may be configured to use any one or more communication technology (e.g., wireless or wired communications) and associated protocols (e.g., Ethernet, Bluetooth®, Bluetooth Low Energy (BLE), Wi-Fi®, WiMAX, 3G LTE, 5G, etc.) to effect such communication.

(19) The telematics system **100** may include multiple vehicles **106** (e.g., a fleet of vehicle **106**), each of which may include an associated telematics device **104**. Each telematics device **104** may be embodied as any type of device capable of performing the functions described herein. For example, the telematics device **104** may be embodied as, without limitation, a mobile computing device, an embedded device, a computer, a laptop computer, a notebook computer, a tablet computer, a wearable computing device, a multiprocessor system, a network appliance, a distributed computing system, a processor-based system, and/or a consumer electronic device. In the illustrative embodiment, each telematics device **104** is a special-purpose computing device configured to collecting and transmitting telematics data, which defines one or more characteristics of the vehicle (e.g., the present location or speed of the vehicle, the type of cargo carried by the vehicle, vehicle operation parameters, vehicle occupancy data, etc.). For example, as discussed below, the telematics device **104** may be configured to be coupled to a communication bus (e.g., a Controller Area Network bus) of the vehicle **106**.

(20) As shown in FIG. 1, the illustrative telematics device **104** includes compute circuitry **140**, an I/O subsystem **146**, a data storage device **148**, a communication subsystem **150**, one or more output devices **152**, a bus connector **154**, and, in some embodiments, local sensors **156**. Of course, the telematics device **104** may include other or additional components, such as those commonly found in a portable or embedded computer (e.g., various input/output devices), in other embodiments. Additionally, in some embodiments, one or more of the illustrative components may be incorporated in, or otherwise form a portion of, another component.

(21) The compute circuitry **140** may be embodied as any type of device or collection of devices capable of performing various compute functions. In some embodiments, the compute circuitry **140** may be embodied as a single device such as an integrated circuit, an embedded system, a field-programmable-array (FPGA), a system-on-a-chip (SOC), or other integrated system or device. Additionally, in some embodiments, the compute circuitry **140** includes or is embodied as a processor **142** and memory **144**. The processor **142** may be embodied as any type of processor capable of performing the functions described herein. For example, the processor **142** may be embodied as a single or multi-core processor(s), digital signal processor, microcontroller, or other processor or processing/controlling circuit. Similarly, the memory **144** may be embodied as any type of volatile or non-volatile memory or data storage capable of performing the functions described herein. In operation, the memory **144** may store various data and software used during operation of the telematics device **104** such as operating systems, applications, programs, libraries, and drivers.

(22) The compute circuitry **140** is communicatively coupled to other components of the telematics

device **104** via the I/O subsystem **146**, which may be embodied as circuitry and/or components to facilitate input/output operations with compute circuitry **140** (e.g., with the processor **142** and/or memory **144**) and other components of the telematics device **104**. For example, the I/O subsystem **146** may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations. In some embodiments, the I/O subsystem **146** may be incorporated, along with the processor **142**, the memory **144**, and other components of the telematics device **104**, into the compute circuitry **140**.

(23) The data storage device **148** may be embodied as any type of device or devices configured for short-term or long-term storage of data such as, for example, memory devices and circuits, memory cards, hard disk drives, solid-state drives, or other data storage devices. The communication subsystem **150** of the telematics device **104** may be embodied as any communication circuit, device, or collection thereof, capable of enabling communications between the telematics device **104**, the telematics server **102**, and/or other remote devices. The communication subsystem **150** may be configured to use any one or more communication technology (e.g., wireless or wired communications) and associated protocols (e.g., Ethernet, Bluetooth®, Bluetooth Low Energy (BLE), WiMAX, 3G LTE, 5G, etc.) to effect such communication.

(24) The output device(s) **152** may be embodied as any type of output device capable of presenting an output or information to a driver of the vehicle **106**. For example, the output device **152** may be embodied as, or otherwise include, a visual output device such as a display screen, an audible output device such as a speaker or buzzer, and/or a tactile output device such as a vibration transducer.

(25) The bus connector **154** may be embodied as any type of vehicle diagnostic connector capable of interfacing with a vehicle bus **160** of the vehicle **106** to receive telemetry data from the components of the vehicle **106** as discussed below. For example, in the illustrative embodiment, the bus connector **154** is embodied as a vehicle diagnostic connector configured to mate with an OBD-II port of the vehicle **106**. In other embodiments, however, the telematics device **104** may be directly connected or otherwise coupled to the vehicle bus **160** with a wiring harness or other connector/connections.

(26) The local sensor(s) **156** may be embodied as any type of sensors located on the telematics device **104** and capable of producing telematics data. For example, the local sensors **156** may be embodied as, or otherwise include, location sensors, accelerometers, gyroscopes, temperature sensors, and/or other sensors.

(27) As shown, each telematics device **104** is coupled to or otherwise included in a corresponding vehicle **106**. Each vehicle **106** may be embodied as, without limitation, a heavy truck, a fleet vehicle, a garbage truck, a snow plow, a dump truck, a bus, a light truck, a passenger car, an airplane, a water craft, or other vehicle. Additionally or alternatively, in some embodiments, the telematics device **104** may be coupled to another asset such as a shipping pallet, a tool cage, a restricted access room, a section of the warehouse, or any other device or location that may be monitored or tracked using a telematics device **104**. Illustratively, the vehicle **106** is a vehicle, and the telematics device **104** is configured to monitor operating conditions or characteristics of the vehicle **106** and to communicate telematics data indicative of the operating conditions of the vehicle **106** to the telematics server **102**.

(28) As discussed above, the illustrative vehicle **106** further includes the vehicle bus **160**, which may be coupled to multiple connected devices such as controllers **162** and sensors **164** of the vehicle **106**. The vehicle bus **160** may be embodied as a controller area network (CAN) bus, a local area network, a wireless network, or another communications network that allows various components of the vehicle **106** and/or peripheral devices to communicate. As discussed above, the telematics device **104** may be coupled to the vehicle bus **160** via the bus connector **154** or via other

mechanisms (e.g., a direct connection).

(29) Each of the controllers **162** may be embodied as an electronic control unit (ECU), an engine controller, a vehicle controller, a microcontroller, or other embedded computing resource of the vehicle **106**. Each controller **162** may provide engine telemetry, ignition signals, odometer signals, or other vehicle telemetry data over the vehicle bus **160**. Each of the sensors **164** may be embodied as a location sensor (e.g., a GPS receiver), a speed sensor, a temperature sensor, an environmental sensor, a weight sensor, a vehicle- or application-specific sensor such as a snow plow position sensor, a fork lift position sensor, a tire pressure sensor, a door state sensor, or other sensor device configured to provide sensor data over the vehicle bus **132**. Although illustrated in FIG. **1** as being coupled to the telematics device **104** via the vehicle bus **160**, it should be understood that in some embodiments one or more of the controllers **162** and/or the sensors **164** may be coupled directly to the telematics device **104** via Bluetooth Low Energy (BLE), wireless networking (WiFi), a direct serial connection, or other connection.

(30) The notification device **108** may be embodied as any type computation or computer device capable of receiving and providing notifications to a user and performing the functions described herein, including, without limitation, a mobile communication device, a smartphone, a desktop computer, a laptop computer, a tablet computer, a custom-built computer, a server, a rack-mounted server, a blade server, a workstation, a network appliance, a web appliance, a consumer electronic device, a distributed computing system, and/or a multiprocessor system. For example, in some embodiments, the notification device **108** may be embodied as a fleet operation computer capable of managing the fleet of vehicles **106** and performing the functions described herein. In other embodiments, the notification device **108** may be embodied as a mobile communication device, such as a smartphone, carried and used by an end user (e.g., a driver of the vehicle **106**).

Furthermore, in some embodiments, the notification device **108** may be embodied as or otherwise incorporated into the telematics device **104** and/or installed in the vehicle **106** as shown in FIG. **1** in dashed lines. Additionally, in some embodiments, the notification device **108** may be embodied as a “virtual computer” formed from multiple computing devices distributed across the network **110** and operating in a public or private cloud.

(31) As shown in FIG. **1**, the illustrative notification device **108** includes compute circuitry **170**, an I/O subsystem **176**, an output device **178** and a communication subsystem **180**. Of course, the notification device **108** may include other or additional components, such as those commonly found in a computer, smartphone, or other computing device (e.g., various input/output devices), in other embodiments. Additionally, in some embodiments, one or more of the illustrative components may be incorporated in, or otherwise form a portion of, another component.

(32) The compute circuitry **170** may be embodied as any type of device or collection of devices capable of performing various compute functions. In some embodiments, the compute circuitry **170** may be embodied as a single device such as an integrated circuit, an embedded system, a field-programmable-array (FPGA), a system-on-a-chip (SOC), or other integrated system or device. Additionally, in some embodiments, the compute circuitry **170** includes or is embodied as a processor **172** and memory **174**. The processor **172** may be embodied as any type of processor capable of performing the functions described herein. For example, the processor **172** may be embodied as a single or multi-core processor(s), digital signal processor, microcontroller, or other processor or processing/controlling circuit. Similarly, the memory **174** may be embodied as any type of volatile or non-volatile memory or data storage capable of performing the functions described herein. In operation, the memory **174** may store various data and software used during operation of the notification device **108** such as operating systems, applications, programs, libraries, and drivers.

(33) The compute circuitry **170** is communicatively coupled to other components of the notification device **108** via the I/O subsystem **176**, which may be embodied as circuitry and/or components to facilitate input/output operations with compute circuitry **170** (e.g., with the processor **172** and/or

memory **174**) and other components of the notification device **108**. For example, the I/O subsystem **176** may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations. In some embodiments, the I/O subsystem **176** may be incorporated, along with the processor **172**, the memory **174**, and other components of the notification device **108**, into the compute circuitry **170**.

(34) The output device(s) **178** may be embodied as any type of output device capable of presenting an output or information to a user of the notification device **108**. For example, the output device **178** may be embodied as, or otherwise include, a visual output device such as a display screen, an audible output device such as a speaker or buzzer, and/or a tactile output device such as a vibration transducer.

(35) The communication subsystem **180** of the notification device **108** may be embodied as any communication circuit, device, or collection thereof, capable of enabling communications between the notification device **108**, the telematics server **102**, and/or other remote devices. The communication subsystem **180** may be configured to use any one or more communication technology (e.g., wireless or wired communications) and associated protocols (e.g., Ethernet, Bluetooth®, Bluetooth Low Energy (BLE), Wi-Fi®, WiMAX, 3G LTE, 5G, etc.) to effect such communication.

(36) As discussed in more detail below, the telematics server **102** is configured to communicate with the telematics device **104** and the notification device **110** and/or other devices of the telematics system **100** over the network **110**. The network **110** may be embodied as any number of various wired and/or wireless networks. For example, the network **110** may be embodied as, or otherwise include, a wired or wireless local area network (LAN), a wired or wireless wide area network (WAN), a cellular network, and/or a publicly-accessible, global network such as the Internet. As such, the network **110** may include any number of additional devices, such as additional computers, routers, stations, and switches, to facilitate communications among the devices of the system **100**.

(37) Referring now to FIG. 2, in the illustrative embodiment, the telematics server **102** is configured to establish an environment **202** during operation. The illustrative environment **202** includes a device interface **210**, a notification manager **212**, a model manager **214**, a data manager **218** and various data sets including telematics database **220** and an event model **222**. The various components of the environment **202** may be embodied as hardware, firmware, software, or a combination thereof. As such, in some embodiments, one or more of the components of the environment **202** may be embodied as circuitry or a collection of electrical devices (e.g., device interface circuitry **210**, notification manager circuitry **212**, ruleset manager circuitry **214**, data and/or manager circuitry **218**). It should be appreciated that, in such embodiments, one or more of those components may form a portion of the processor **122**, the I/O subsystem **126**, and/or other components of the telematics server **102**.

(38) The device interface **210** is configured to establish communications with the telematics device **104** and the notification device **110**. In use, the device interface **210** receives telematics data from the telematics device **104** and, when appropriate, transmits a notification to the notification device **110** based on an analysis of the received telematics data as discussed below. The notification may be embodied as any type of notification. For example, in some embodiments, the notification may include instructions for the notification device **110**. For example, the notification may instruct the notification device **110** to generate an alert. Additionally or alternatively, the notification may be configured to provide information to the user of the notification device **110** (e.g., information regarding the determined unexpected event), which the notification device **110** may present to the user via the output device **178**.

(39) The notification manager **212** is configured to analyze the telematics data received from the

telematics device **104** and determine whether an unexpected event has occurred based on the event model **222**. That is, the notification manager **212** compares the received telematics data to one or more rules included in or otherwise defined by the event model **222** to determine whether the telematics data define an unexpected (e.g., abnormal) event based on historical telematics data. As discussed in more detail below, the event model **222** includes a number of rules which define an expected operational condition or characteristic of the vehicle **106** (i.e., expected telematics data). For example, the event model **222** may include temporal-based rules, location-based rules, historical based rules, sequential-based rules, relational-based rules, and/or other types of rules that define expected telematics data. In some embodiments, for example, the event model **222** may include a rule that indicates the vehicle **106** is typically at a particular location at a particular time. In such embodiment, if the received telematics data does not indicate that the vehicle **106** is located at the expected location, the notification manager **212** may determine that an unexpected event has occurred and generate an associated notification for transmission to the notification device **108**. Other operational aspects of the vehicle **106** may also be monitored for unexpected events such as sudden changes in the driving pattern of the vehicle **106** such as excessive acceleration, excessive deceleration, and/or lane departures.

(40) The model manager **214** is configured to construct, maintain, and update the event model **222** based on newly received telematics data and, in some embodiments, feedback from the user of the notification device(s) **108**. As discussed above, the event model **222** includes a number of rules that define an expected operational condition or characteristic of the vehicle **106** (i.e., expected telematics data) based on historical telematics data. The model manager **214** may initially define those rules during an initial training period during which telematics data is monitored using an unsupervised machine learning algorithm (e.g., to identify commonality in the telematics data). Over time, the model manager **214** is configured to continually update the event model **222** such that the rules of the event model **222** define the presently expected operational conditions and/or characteristics of the vehicle **106**. To do so, in some embodiments, the model manager **214** may include a machine learning engine **216** configured to monitor the received telematics data overtime and update the event model **222** based on “learned” operational conditions and/or characteristics of the vehicle **106**. In this way, the event model **222** can be maintained in a current form without manual updating. Additionally, the user of the notification device **108** may provide feedback based on generated notifications, which the model manager **214** may use to further update the event model **222** using a type of supervised machine learning algorithm.

(41) The data manager **218** is configured to manage the telematics data received from the various telematics devices **104**. For example, the data manager **218** stores the received telematics data in the telematics database **220**. To do so, the data manager **218** may utilize any suitable storage paradigm.

(42) Still referring to FIG. 2, in the illustrative embodiment, the telematics device **104** establishes an environment **204** during operation. The illustrative environment **204** includes a data collection manager **230**, a data transmission manager **232**, a server interface **234**, an alert manager **236** and various data sets including a local telematics database **240** and, in some embodiments, a transmission ruleset **242**. Similar to environment **202**, the various components of the environment **204** may be embodied as hardware, firmware, software, or a combination thereof. As such, in some embodiments, one or more of the components of the environment **204** may be embodied as circuitry or a collection of electrical devices (e.g., data collection manager circuitry **230**, data transmission manager circuitry **232**, server interface circuitry **234**, and/or alert manager circuitry **236**). It should be appreciated that, in such embodiments, one or more of those components may form a portion of the processor **142**, the I/O subsystem **146**, and/or other components of the telematics device **104**.

(43) The data collection manager **230** is configured to collect telematics data from various controllers **162** and/or sensors **164** of the vehicle **106**. As discussed above, in some embodiments,

the telematics device **104** may receive such telematics data via the vehicle bus **160** and bus connector **154**. Additionally, in embodiments in which the telematics device **104** includes the local sensors **156**, the data collection manager **230** may also receive or collect telematics data from the local sensors **156**. In doing so, the data collection manager **230** may utilize any suitable data collection scheme including, for example, data pushes, data pulls, and/or data doorbells. As such, the telematics data may be sampled or received according to any suitable data acquisition scheme.

(44) The data transmission manager **232** is configured to manage the transmission of the telematics data from the telematics device **104** to the telematics server **102**. To do so, the data transmission manager **232** may use any suitable transmission schedule to transfer the telematics data to the telematics server **102**. For example, in some embodiments, the data transmission manager **232** may be transmit the telematics data as it is received from the various controllers **162**, sensors **164**, and/or sensors **156**. In other embodiments, the data transmission manager **232** may transmit the telematics data based on a schedule set forth in the transmission ruleset **242**. That is, the transmission ruleset **242** may define a set of rules that dictate a frequency at which the telematics data should be transmitted. For example, the transmission ruleset **242** may define a batch timing (e.g., every 30 seconds) at which all collected telematics data is to be transmitted, a data threshold amount at which the collected telematics data is to be transmitted, and/or other conditions under which the telematics data should be transmitted to the telematics server **102**. In some embodiments, the transmission rules may be based on the type of telematics data (e.g., important telematics data may be transmitted at a higher frequency than non-important telematics data).

(45) The server interface **234** is configured to establish communications with the telematics server **102**. In use, the server interface **234** transmits the telematics data when instructed to do so by the data transmission manager **232**. Additionally, in some embodiments (e.g., those embodiments in which the telematics device **104** includes or is embodied as the notification device **108**), the server interface **234** may receive notifications transmitted by the telematics server **102** (as well as any other data), based on the transmitted telematics data.

(46) In embodiments in which the telematics device **104** includes or is embodied as the notification device **108**, the alert manager **236** is configured to manage the notifications received from the telematics server **102**. In embodiments in which the notification includes instructions, the alert manager **236** is configured to execute the instructions of the received notification. For example, the alert manager **236** may activate an alert device of the telematics device **104** and/or vehicle **106** based on the notification. Additionally or alternatively, in embodiments in which the notification includes information to be provided to the driver, the alert manager **236** is configured to present the information to the driver via the output device **152** (e.g., via a display screen or speaker).

(47) In some embodiments, the alert manager **236** may include a local event model **238**. The local event model **238** may be a copy of the event model **222**, or a subset thereof, and is received by the telematics device **104** from the telematics server **102**. In such embodiments, the alert manager **236** may be further configured to operate in a manner similar to the notification manager **212** of the telematics server **102**. That is, the alert manager **236** may locally analyze the telematics data to determine whether an unexpected event has occurred based on the local event model **238** similar to the functionality of the notification manager **212** discussed above.

(48) Continuing reference to FIG. 2, in the illustrative embodiment, the notification device **108** establishes an environment **206** during operation. The illustrative environment **206** includes a communication manager **250** and an alert manager **252**. Similar to environments **202** and **204**, the various components of the environment **206** may be embodied as hardware, firmware, software, or a combination thereof. As such, in some embodiments, one or more of the components of the environment **206** may be embodied as circuitry or a collection of electrical devices (e.g., communication manager circuitry **250** and/or notification manager circuitry **252**). It should be appreciated that, in such embodiments, one or more of those components may form a portion of the processor **172**, the I/O subsystem **176**, and/or other components of the notification device **108**.

(49) The communication manager **250** is configured to manage communications of the notification device **108**. In use, the communication manager **250** receives notifications from the telematics server **102**. Additionally, in some embodiments as discussed in more detail below, the communication manager **250** may transmit user feedback to the telematics server **102**. Such feedback may be based on the received notifications.

(50) The alert manager **252** is configured to manage the notifications received from the telematics server **102**. In embodiments in which the notification includes instructions, the alert manager **252** is configured to execute the instructions of the received notification. For example, in embodiments in which the notification includes information to be provided to the user, the alert manager **252** is configured to present the information to the user via the output device **178** (e.g., via a display screen or speaker). Additionally, in embodiments in which the notification includes instructions, the alert manager **252** may activate an alert device (e.g., a visual, tactile, or audible alert device) of the notification device **108** based on the notification.

(51) Referring now to FIG. **3**, in use, the telematics device **104** may execute a method **400** for managing telematics data. It should be appreciated that, in some embodiments, the operations of the method **300** may be performed by one or more components of the environment **204** of the telematics device **104** as shown in FIG. **2**.

(52) The method **300** begins with block **302**, in which the telematics device **104** may perform various initializations procedures. For example, the telematics device **104** may load an operating system and/or various software or software extensions in block **302** as part of a power-up cycle. Subsequently, in block **304**, the telematics device **104** receives telematics data. For example, in block **306**, the telematics device **104** may receive the telematics data from one or more vehicle sensors **164**. Additionally or alternatively, in block **308**, the telematics device **104** may receive the telematics data from one or more vehicle controllers **162**. Furthermore, in embodiments in which the telematics device **104** includes the local sensors **156**, the telematics device **104** may receive telematics data from the local sensors **156** in block **310**.

(53) In block **312**, the telematics device stores the received telematics data in the local database **240**. In doing so, in some embodiments, the telematics device **104** may compact the telematics data in block **314**. For example, the telematics device **104** may use a suitable compression algorithm or saving schema to reduce the overall size of the saved telematics data. Regardless, in block **316**, the telematics device **104** transmits the newly received telematics data to the telematics server **102**. In doing so and as discussed above, the telematics device **104** may be configured to transmit the telematics data as it is received from the sensors **164**, controllers **162**, and/or local sensors **156**. Alternatively, in other embodiments in block **318**, the telematics server **102** may be configured to transmit the telematics data based on a schedule set forth in the transmission ruleset **242** as discussed above.

(54) Referring now to FIG. **4**, in use, the telematics server **102** may execute a method **400** for constructing the event model **222** to facilitate analyzing telematics data. It should be appreciated that, in some embodiments, the operations of the method **400** may be performed by one or more components of the environment **202** of the telematics server **102** as shown in FIG. **2**.

(55) The method **400** begins with block **402**, in which the telematics server **102** may perform various initializations procedures. For example, the telematics server **102** may load an operating system and/or various software or software extensions in block **402** as part of a power-up cycle. In some embodiments, in block **404**, the telematics server **102** may retrieve, receive, or otherwise obtain a pre-trained event model **222**. The pre-trained event model **222** may include a number of predefined or pre-established rules, including predefined temporal-based rules, predefined location-based rules, predefined historical-based rules, pre-defined sequential-based rules, predefined relational-based rules, and/or other rules and/or conditions. Such predefined rules may be manually defined in the pre-trained event model **222** or may be constructed during a training session using training telematics data. For example, the pre-trained event model **222** may define “global” rules

that are applicable to a whole set or range of telematics devices **104** or otherwise considered universal.

(56) After the telematics server **102** has performed the various initialization procedures in block **402**, the telematics server **102** monitors for telematics data from the telematics device **104** in block **406**. If telematics data is received in block **506**, the method **400** advances to block **408** in which the telematics server **102** stores the received telematics data in the local telematics database **220**.

(57) Subsequently, in block **410**, the model manager **214** of the telematics server **102** performs unsupervised learning on the received and stored telematics data using the machine learning engine **216**. The learning is unsupervised in that the telematics data is unlabeled from the driver of the vehicle **106** or other user. In performing the unsupervised learning on the received telematics data, the telematics server defines one or more rules of the event model **222**. For example, in block **412**, the telematics server **102** may define one or more temporal-based rules. Each temporal-based rule may define an expected temporal operational condition and/or characteristic of the vehicle **106**. For example, a temporal-based rule may define an expected time period of operation of the vehicle **106**, an expected time at which the vehicle **106** should be started or stopped, an expected time at which the vehicle **106** should be located at a particular location, an expected completion time of a trip, and/or other time-based operational conditions and/or characteristics. In block **414**, the telematics server **102** may define one or more location-based rules. Each location-based rule may define an expected location-related operational condition and/or characteristic of the vehicle **106**. For example, a locational-based rule may define a geofenced area of operation for the vehicle **106**, an expected location of the vehicle **106**, an expected route of the vehicle **106**, and/or other location-based operational conditions and/or characteristics.

(58) In block **416**, the telematics server **102** may define one or more historical-based rules. Each historical-based rule may define an expected operational condition and/or characteristic of the vehicle **106** given the past telematics data. For example, a historical based rule may define an expected maintenance of the vehicle **106** based on a previous maintenance of the vehicle **106**, an expected speed (or max speed) of the vehicle **106** based on a previous speed or location of the vehicle **106**, and/or other historical-based operational conditions and/or characteristics. In block **418**, the telematics server **102** may define one or more sequential-based rules. Each sequential-based rule may define an expected operational condition and/or characteristic of the vehicle **106** given an immediate past operational condition and/or characteristic of the vehicle **106**. For example a sequential-based rule may define an expected location of the vehicle **106** based on the immediately prior location of the vehicle **106**, an expected weight of the vehicle based on the immediately prior location of the vehicle **106**, and/or other historical-based operational conditions and/or characteristics. In block **420**, the telematics server **102** may manually define one or more relational-based rules. Each relational-based rule may define an expected operational condition and/or characteristic of the vehicle **106** given the existence of another operational condition and/or characteristic of the vehicle **106**. For example, a relational-based rule may define an expected cargo of the vehicle **106** given a present location of the vehicle **106**, an expected max speed of the vehicle **106** given a present location of the vehicle **106**, an expected operation start time given the particular day of the week, and/or other relational-based operational conditions and/or characteristics. In some embodiments, a single rule of the event model **222** may be composed of multiple sub-rules. For example, a rule may include both a location-based rule and a temporal-based rule (e.g., a rule that defines an expected location at a particular time of the day). It should be appreciated, however, that the telematics server **102** may define additional or other rules during the unsupervised learning based on the observed data by identifying patterns and/or outliers in the received telematics data.

(59) During or subsequent to the unsupervised learning of block **410**, the telematics server **102** constructs or otherwise generates the event model **222** based on the various rules defined during such learning. In some embodiments, the telematics server **102** may also perform supervised

training in which a user of the telematics server **102** may “label” the received telematics data to improve the creation of the initial set of event rules. Additionally, in embodiments in which a pre-trained event model **222** is used, the telematics server **102** may update the pre-trained model to construct or generate the final event model **222**. Regardless, in block **426**, the telematics server **102** subsequently saves the generated event model **222**.

(60) Referring now to FIG. **5**, in use, the telematics server **102** may execute a method **500** for providing notifications to the notifications device **108**. It should be appreciated that, in some embodiments, the operations of the method **500** may be performed by one or more components of the environment **202** of the telematics server **102** as shown in FIG. **2**.

(61) The method **500** begins with block **502**, in which the telematics server **102** may perform various initializations procedures. For example, the telematics server **102** may load an operating system and/or various software or software extensions in block **502** as part of a power-up cycle. Subsequently, in block **504**, the telematics server **102** monitors for telematics data from the telematics device **104**. If telematics data is received in block **504**, the method **500** advances to block **506** in which the telematics server **102** stores the received telematics data in the local telematics database **220**.

(62) Subsequently, in block **508**, the telematics server determines whether an unexpected event has occurred based on the received telematics data and the event model **222**. As discussed above, the event model **222** includes or defines a number of rules that define expected operational conditions and/or characteristics of the vehicle **106**. If the received telematics data does not match the relevant event rule, the telematics server **102** determines that an unexpected event (e.g., unexpected telematics data) has occurred. As such, the telematics server **102** may compare the received telematics data to one or more rules of the event model **222**. For example, the telematics server **102** may compare the received telematics data to one or more temporal-based rules in block **510**, to one or more location-based rules in block **512**, to one or more historical-based rules in block **514**, to one or more sequential-based rules in block **516**, to one or more relational-based rules in block **518**, and/or to one or more additional rules.

(63) In block **520**, the telematics server **102** determines whether an unexpected event has been detected based on the analysis performed in block **508**. If so, the method **500** advances to block **522** in which the telematics server **102** generates and transmits a notification to the notification device **108**. As discussed above, the notification may be embodied as a simple alert notification and/or may provide instructions to the telematics device **104** for providing information to the driver of the vehicle **106**.

(64) If no unexpected event was detected in block **520** or after the notification has been transmitted in block **522**, the method **500** advances to block **524**. In block **524**, the telematics server **102** determines whether to update the event model **222**. In some embodiments, the telematics server **102** may update the event model **222** based on every received telematics data. In other embodiments, the telematics server **102** may be configured to periodically update the event model **222** or update the event model **222** after a particular amount of telematics data has been received. Additionally, if the user of the notification device **108** has provided feedback in response to a notification, the telematics server **102** may update the event model **222** based on such feedback so as to perform a type of supervised machine learning using the feedback as labeled data. Regardless, if the telematics server **102** determines to update the telematics data in block **524**, the method **500** advances to block **526** in which the telematics server **102** updates the event model **222** based on the recently received telematics data and/or user feedback. For example, in the illustrative embodiment in block **528**, the telematics server **102** updates the event model **222** based on the received telematics data and/or user feedback using the machine learning engine **216** (e.g., by using unsupervised or supervised learning, respectively). In doing so, the machine learning engine **216** may use any suitable machine learning algorithm to update the event model **222**. In some embodiments, in block **530**, the telematics server **102** may also transmit the updated event model

222 to the telematics device **104**. In such embodiments, the telematics device **104** may utilize the updated event model **222** to perform some amount of local analysis of the telematics data as discussed above.

(65) Referring now to FIG. **6**, in use, the notification device **108** may execute a method **600** for managing notifications received from the telematics server **102**. As discussed above, the notification device **108** may be embodied as a separate computer device (e.g., a fleet manager server, a smartphone or mobile communication device, etc.) in some embodiments. In other embodiments, the notification device **108** may be incorporated into or otherwise embodied as the telematics device **104**. In such embodiments, the telematics device **104** may execute the method **600**. Regardless, it should be appreciated that the operations of the method **400** may be performed by one or more components of the environment **206** of the notification device **108** as shown in FIG. **2**.

(66) The method **600** begins with block **602**, in which the notification device **108** may perform various initializations procedures. For example, the notification device **108** may load an operating system and/or various software or software extensions in block **402** as part of a power-up cycle. Subsequently, in block **604**, the notification device **108** monitors for notifications received from the telematics server **102**. If a notification is received, the method **600** advances to block **606**.

(67) In block **606**, the notification device **108** generates an alert based on the received notification. For example, in block **608**, the notification device **108** may activate an output device **178** to alert the user of the notification device **108** about the notification. The particular output device **178** used may be determined based on the notification and/or information included therein. As such, the alert may be embodied as a simple audible, visual, or tactile alert in embodiments in which the notification is a simple alert notification. However, in other embodiments in which the notification includes information, the alert may be embodied as the presentation of such information to the user via a respective output device **178** (e.g., via a speaker or display). Additionally or alternatively, in some embodiments, the notification device **108** may transmit the alert to other devices. For example, in embodiments in which the notification device **108** is located within the vehicle **106** or otherwise associated with the vehicle **106**, the notification device **108** may be configured to transmit the alert to the telematics device **104**, which may in turn generate the alert via one of the output devices **152**.

(68) Subsequently, in block **612**, the notification device **108** determines whether to obtain user feedback regarding the received notification. For example, in some embodiments, the notification device **108** may request user feedback for each notification or request feedback only in response to particular criteria (e.g., the number of notifications received within a reference time frame). Additionally or alternatively, the user may manually provide feedback as desired. Regardless, it should be appreciated that the user feedback embodies a type of labeling of the notification, which may be used by the telematics server **102** to update the event model **222** using a supervised machine learning algorithm as discussed above.

(69) If the notification device **108** determines to obtain user feedback in block **612**, the method **600** advances to block **614** in which the notification device **108** prompts the user for feedback on the recently received notification. The notification device **108** subsequently receives the user feedback in block **616** and transmits the feedback to the telematics server **102** in block **618**. The method **600** subsequently loops back to block **604** in which the notification device **108** continues to monitor for notifications from the telematics server **102**.

(70) Referring now to FIG. **7**, in embodiments in which the telematics device **104** is configured to perform the local analysis of the telematics data and includes the local event model **238**, the telematics device **104** may execute a method **700** for managing telematics data. It should be appreciated that, in some embodiments, the operations of the method **700** may be performed by one or more components of the environment **204** of the telematics device **104** as shown in FIG. **2**.

(71) The method **700** begins with block **702**, in which the telematics device **104** determines

whether an update to the local event model **238** is available from the telematics server **102**. If so, the method **700** advances to block **704** in which the telematics device **104** receives the updated event model **238** from the telematics server **104**. Subsequently, in block **706**, the telematics device **104** receives telematics data. For example, in block **708**, the telematics device **104** may receive the telematics data from one or more vehicle sensors **164**. Additionally or alternatively, in block **710**, the telematics device **104** may receive the telematics data from one or more vehicle controllers **162**. Additionally, in embodiments in which the telematics device **104** includes the local sensors **156**, the telematics device **104** may receive telematics data from the local sensors **156** in block **712**.

(72) In block **714**, the telematics device stores the received telematics data in the local database **240**. In doing so, in some embodiments, the telematics device **104** may compact the telematics data in block **716** as discussed above. Subsequently, in block **718**, the telematics device **104** transmits the newly received telematics data to the telematics server **102**. In doing so and as discussed above, the telematics device **104** may be configured to transmit the telematics data as it is received from the sensors **164**, controllers **162**, and/or local sensors **156**. Alternatively, in other embodiments in block **720**, the telematics server **102** may be configured to transmit the telematics data based on a schedule set forth in the transmission ruleset **242** as discussed above.

(73) Subsequently, in block **622**, the telematics device **104** determines whether an unexpected event has occurred based on an analysis of the received telematics data and the local event model **238**. To do so, the telematics device **104** may compare the received telematics data to one or more rules of the local event model **238** using a methodology similar to the telematics server **102** described above in regard to block **508** of method **500**.

(74) In block **724**, the telematics device **104** determines whether an unexpected event has been detected based on the analysis performed in block **722**. If so, the method **700** advances to block **726** in which the telematics device **104** generates an alert to the driver of the vehicle **106**. As discussed above, the alert may be embodied as a simple alert notification and/or may provide information to the driver of the vehicle **106**. Additionally, in some embodiments in block **728**, the telematics device **104** may receive feedback from the driver of the vehicle **106** based on the generated alert and update the local event model **238** based on the feedback using a supervised machine learning algorithm as discussed above. Regardless, the method **700** subsequently loops back to block **702** in which the telematics device **104** continues to monitor for an updated event ruleset from the telematics server **102**.

(75) While the disclosure has been illustrated and described in detail in the drawings and foregoing description, such an illustration and description is to be considered as illustrative and not restrictive in character, it being understood that only illustrative embodiments have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

(76) There are a plurality of advantages of the present disclosure arising from the various features of the methods, apparatuses, and systems described herein. It will be noted that alternative embodiments of the methods, apparatuses, and systems of the present disclosure may not include all of the features described yet still benefit from at least some of the advantages of such features. Those of ordinary skill in the art may readily devise their own implementations of the methods, apparatuses, and systems that incorporate one or more of the features of the present invention and fall within the spirit and scope of the present disclosure as defined by the appended claims.

Claims

1. A telematics server for providing notifications to a telematics device of a vehicle, the telematics server comprising: an event model comprising a plurality of rules that each define an expected event associated with the vehicle, wherein the expected event defines at least one expected characteristic of the vehicle based on historical telematics data of the vehicle; a device interface

configured to receive telematics data from the telematics device, wherein the telematics data defines at least one present characteristic of the vehicle and is indicative of an event associated with the vehicle; and a notification manager configured to determine whether the event is an expected event or an unexpected event based on the event model, wherein to determine whether the event is an expected event or an unexpected event comprises to (i) compare the telematics data to a rule of the plurality of rules, (ii) determine whether the at least one present characteristic of the vehicle matches the at least one expected characteristic of the vehicle defined by the expected event defined by the rule, (iii) determine that the event is an expected event in response to a determination that the at least one present characteristic matches the at least one expected characteristic of the event; and (iii) determine that the event is an unexpected event in response to a determination that the at least one present characteristic does not match the at least one expected characteristic of the vehicle, and wherein the device interface is further configured to transmit an event notification to a notification device different from the telematics device in response to a determination that the event is an unexpected event and to not transmit an event notification to the notification device in response to a determination that the event is an expected event such that the telematics server is configured to transmit the event notification to the notification device only if the event is an unexpected event.

2. The telematics server of claim 1, wherein the notification manager is configured to update the event model based on the telematics data received from the telematics device or based on feedback received from the notification device using a machine learning engine.

3. The telematics server of claim 1, wherein the notification instructs the notification device to generate an alert.

4. The telematics server of claim 1, wherein the notification is configured to provide information related to the unexpected event to a user of the notification device.

5. The telematics server of claim 1, wherein the telematics server further includes a model manager comprising a machine learning engine configured to generate the event model based on the historical telematics data using unsupervised learning.

6. The telematics server of claim 1, wherein to analyze the telematics data comprises to analyze the telematics data based on a temporal-based rule of the event model, wherein the temporal-based rule defines an expected temporal event related to the vehicle based on the historical telematics data.

7. The telematics server of claim 1, wherein to compare the telematics data comprises to compare the telematics data to a location-based rule of the event model, wherein the location-based rule defines an expected location of the vehicle based on historical telematics data.

8. The telematics server of claim 1, wherein to compare the telematics data comprises to compare the telematics data to a historical-based rule of the event model, wherein the historical based rule defines an expected activity related to the vehicle based on the historical telematics data.

9. The telematics server of claim 1, wherein to compare the telematics data comprises to compare the telematics data to a sequential-based rule of the event model, wherein the sequential-based rule defines an expected event related to the vehicle given an occurrence of a prior event related to the vehicle.

10. The telematics server of claim 1, wherein to compare the telematics data comprises to compare the telematics data to a relational-based rule of the event model, wherein the relational-based rule defines an expected characteristic of the vehicle given a state of another characteristic of the vehicle.

11. A method for providing notifications to a telematics device of a vehicle, the method comprising: receiving, by a telematics server, telematics data from the telematics device, wherein the telematics data defines one or more characteristics of the vehicle and is indicative of an event associated with the vehicle; analyzing, by the telematics server, the telematics data using an event model managed by the telematics server to determine whether the event is an expected event or an unexpected event based on the event model, wherein the event model includes a plurality of rules that each define an

expected event associated with the vehicle and the expected event defines at least one expected characteristic of the vehicle based on historical telematics data, and wherein analyzing the telematics data comprises: (i) comparing the telematics data to a rule of the plurality of rules, (ii) determining whether the at least one present characteristic of the vehicle matches the at least one expected characteristic of the vehicle defined by the expected event defined by the rule, (iii) determining that the event is an expected event in response to a determination that the at least one present characteristic matches the at least one expected characteristic of the event, and (iv) determining that the event is an unexpected event in response to a determination that the at least one present characteristic does not match the at least one expected characteristic of the vehicle; transmitting, by the telematics server, an event notification to a notification device different from the telematics device in response to a determination that the event is an unexpected event; and forgoing transmission of an event notification to the notification device in response to a determination that the event is an expected event such that the telematics server transmits the event notification to the notification device only if the event is an unexpected event.

12. The method of claim 11, further comprising updating, by the telematics server, the event model based on the telematics data received from the telematics device or based on feedback received from the notification device using a machine learning engine.

13. The method of claim 11, wherein the notification instructs the notification device to generate an alert or is configured to provide information related to the unexpected event to a user of the notification device.

14. The method of claim 11, wherein analyzing the telematics data comprises analyzing the telematics data based on a temporal-based rule of the event model, wherein the temporal-based rule defines an expected temporal event related to the vehicle based on the historical telematics data.

15. The method of claim 11, wherein analyzing the telematics data comprises analyzing the telematics data based on a location-based rule of the event model, wherein the location-based rule defines an expected location of the vehicle based on historical telematics data.

16. The method of claim 11, wherein analyzing the telematics data comprises analyzing the telematics data based on a historical-based rule of the event model, wherein the historical based rule defines an expected activity related to the vehicle based on the historical telematics data.

17. The method of claim 11, wherein analyzing the telematics data comprises analyzing the telematics data based on a sequential-based rule of the event model, wherein the sequential-based rule defines an expected event related to the vehicle given an occurrence of a prior event related to the vehicle.

18. The method of claim 11, wherein analyzing the telematics data comprises analyzing the telematics data based on a relational-based rule of the event model, wherein the relational-based rule defines an expected characteristic of the vehicle given a state of another characteristic of the vehicle.

19. One or more non-transitory, computer-readable storage media comprising a plurality of instructions that, in response to execution, cause a telematics server to: receive telematics data from the telematics device, wherein the telematics data defines one or more characteristics of the vehicle and is indicative of an event associated with the vehicle; analyze the telematics data using an event model managed by the telematics to determine whether the event is an expected event or an unexpected event based on the event model, wherein the event model includes a plurality of rules that each define an expected event associated with the vehicle and the expected event defines at least one expected characteristic of the vehicle based on historical telematics data, and wherein analyzing the telematics data comprises to: (i) compare the telematics data to a rule of the plurality of rules, (ii) determine whether the at least one present characteristic of the vehicle matches the at least one expected characteristic of the vehicle defined by the expected event defined by the rule, (iii) determine that the event is an expected event in response to a determination that the at least one present characteristic matches the at least one expected characteristic of the event, and (iv)

determine that the event is an unexpected event in response to a determination that the at least one present characteristic does not match the at least one expected characteristic of the vehicle; transmit an event notification to a notification device different from the telematics device in response to a determination that the event is an unexpected event; and forego transmission of an event notification to the notification device in response to a determination that the event is an expected event such that the telematics server is configured to transmit the event notification to the notification device only if the event is an unexpected event.

20. The one or more non-transitory, computer-readable storage media of claim 19, wherein the plurality of instructions, in response to execution, further cause the telematics server to update the event model based on the telematics data received from the telematics device or based on feedback received from the notification device using a machine learning engine.
